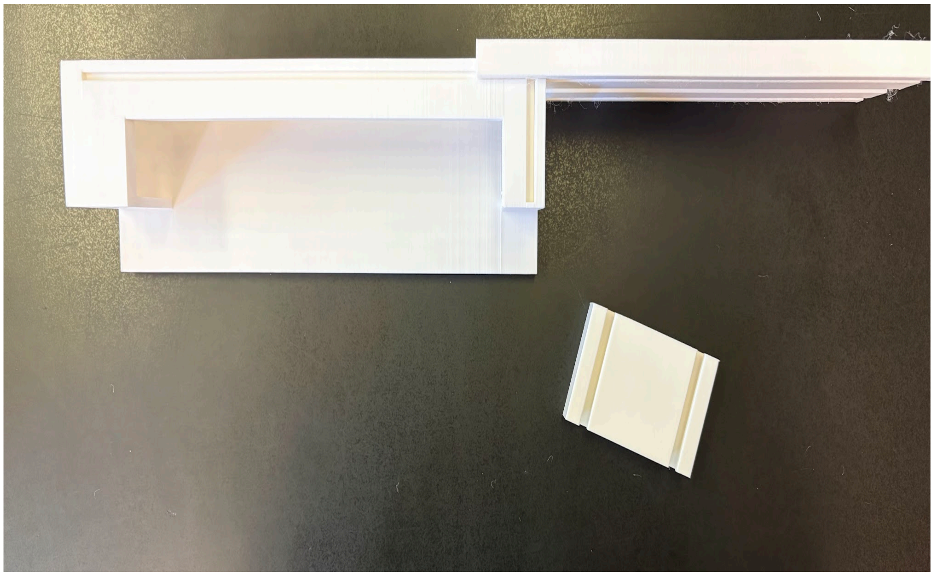
Element D: Design Concept Generation, Analysis, and Selection

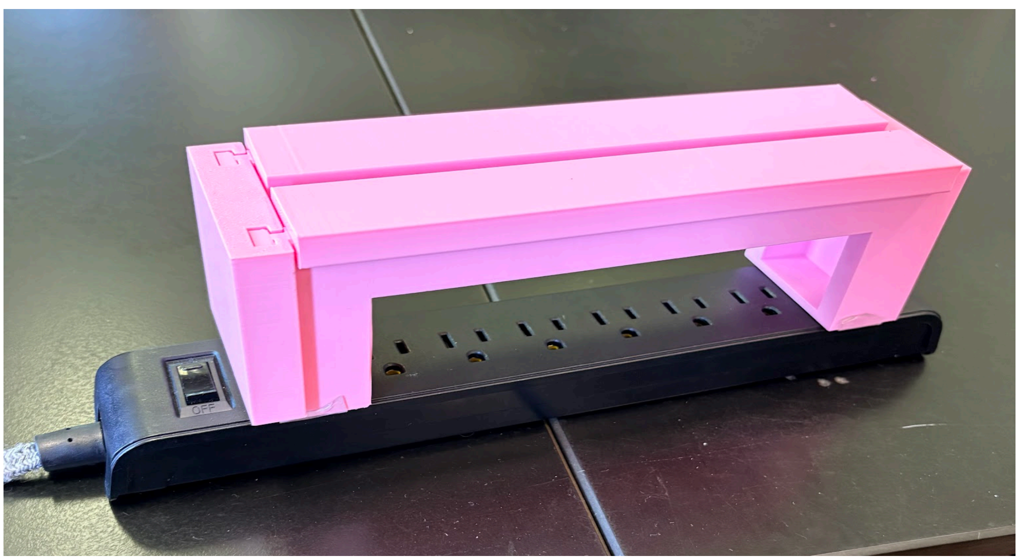
Power cables often come unplugged accidentally for many reasons, like getting kicked or pulled from their socket unintentionally. This affects the general public. The goal of our project is to prevent cables from becoming unplugged by accidentally getting kicked or pulled on. This will help people prevent loss of computer data, loss of aborted creations when 3D printing, or loss of electricity to refrigerators and freezers.

We brainstormed in our notebooks for 10 minutes on our own, then shared our ideas with our team and wrote down any ideas we didn't have. Then we gave feedback on each other's ideas and narrowed them down to five ideas and roughly sketched them. We used a decision matrix to compare our current solution, which is crawling under a table to plug cables back in, to a 3D printed power strip cover, a wall plug clip, velcro, magnets, and a spiral cable. Our decision matrix compared them by how safe they are to use, if it permanently damages outlets, if they cost less than $.50 per outlet or cable secured, if it was durable enough for at least five installments, if they can be used on a wide variety of outlets and cables, and if they can be installed in under 60 seconds. We weighed safety and cost higher because we thought they were the most important.

(Prototype 1 with sliders partly attached)



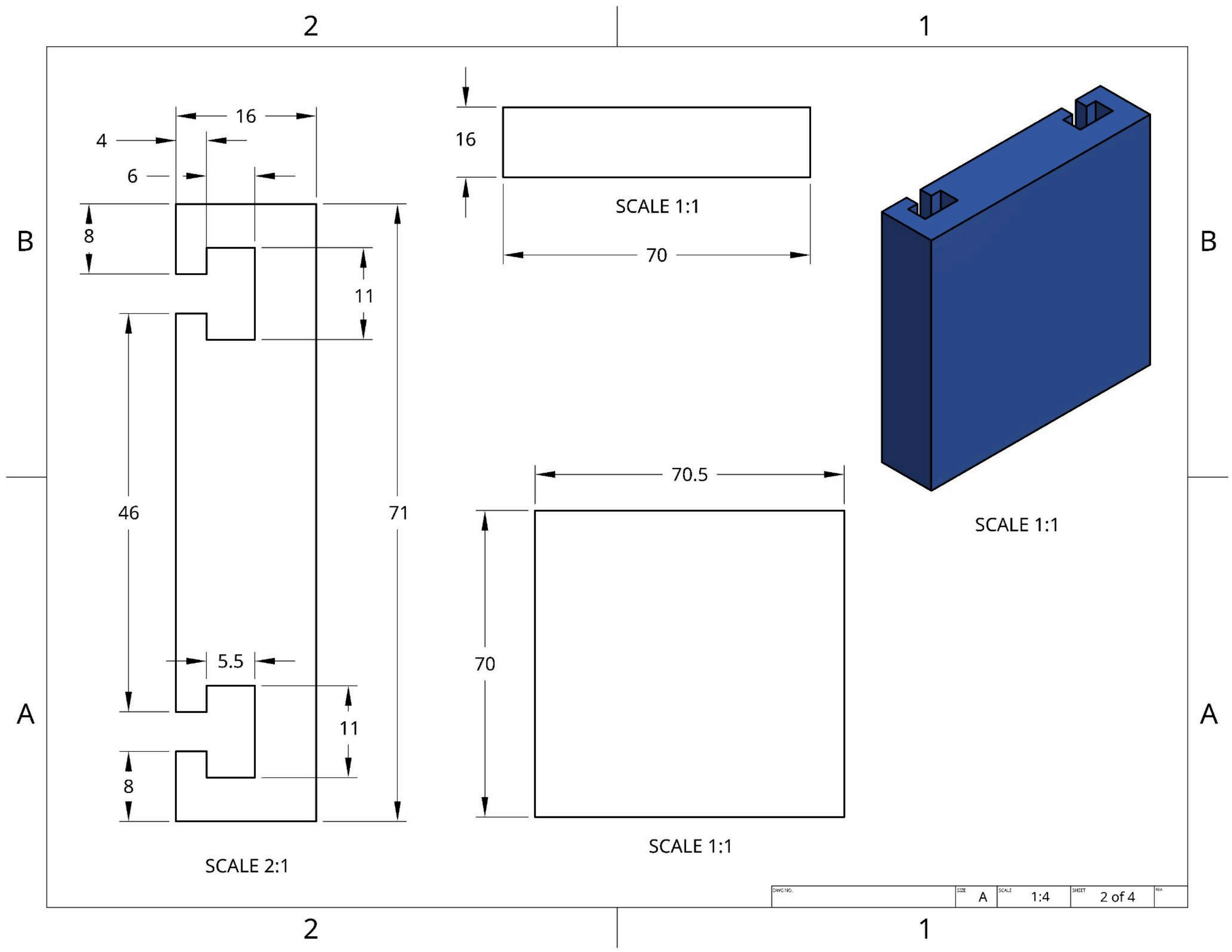
(prototype 2 installed on power strip)



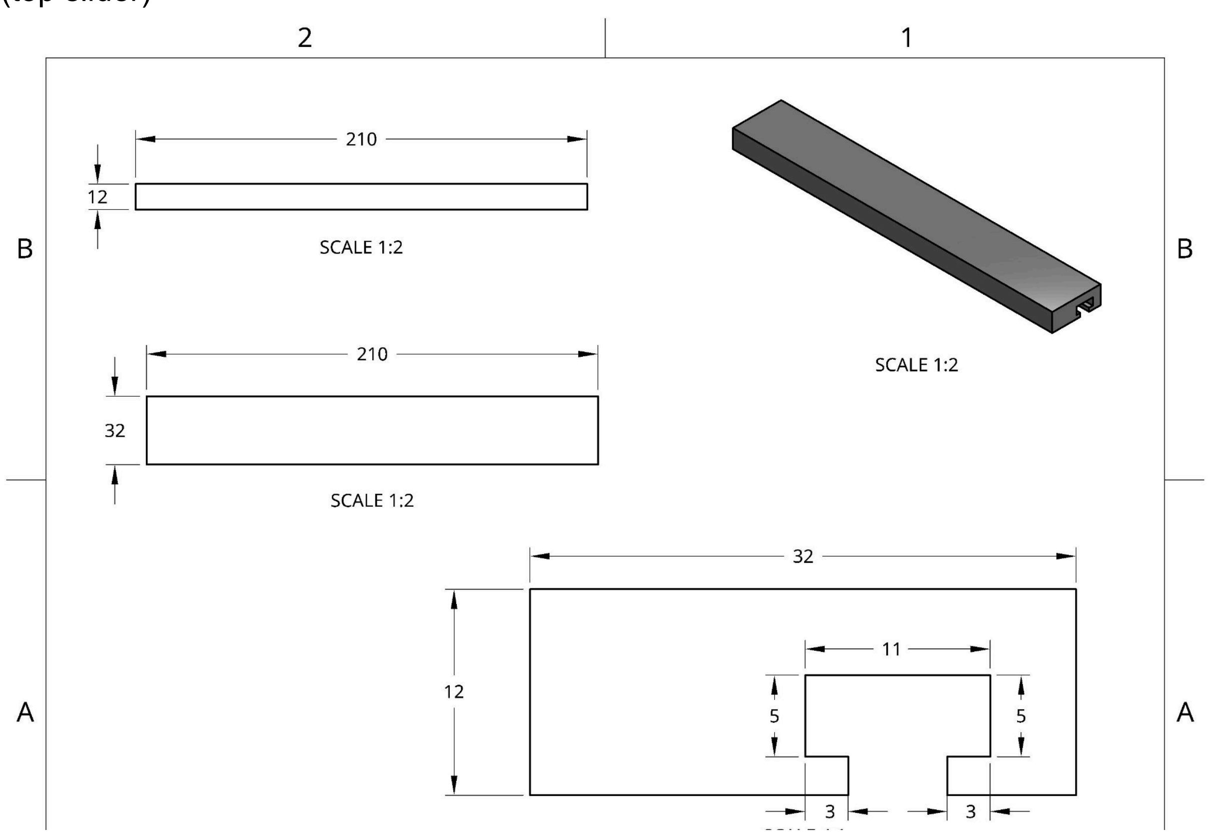
Our final solution is a 3D printed box that goes over a power strip. The box stops the cables from being unplugged. There were other ideas with more points in our design matrix, but we did not include them because it was the most realistic for us to make in class with the available time.

To build our prototype, we started by designing it in Onshape, a CAD modeling software. We sent the sketch to the 3D printers for printing. Then we assembled the parts we printed and added double-sided tape to the bottom of the cable holder so we could attach it to the power strip. To make this, you would need a 3D printer, PLA (polylactic acid) filament, double-sided adhesive, scissors, and a computer with CAD modeling software. The estimated cost is about five dollars and 50 cents per solution because it takes about a quarter roll of filament, which is 5 dollars and 4 inches of tape, which is about 15 cents.

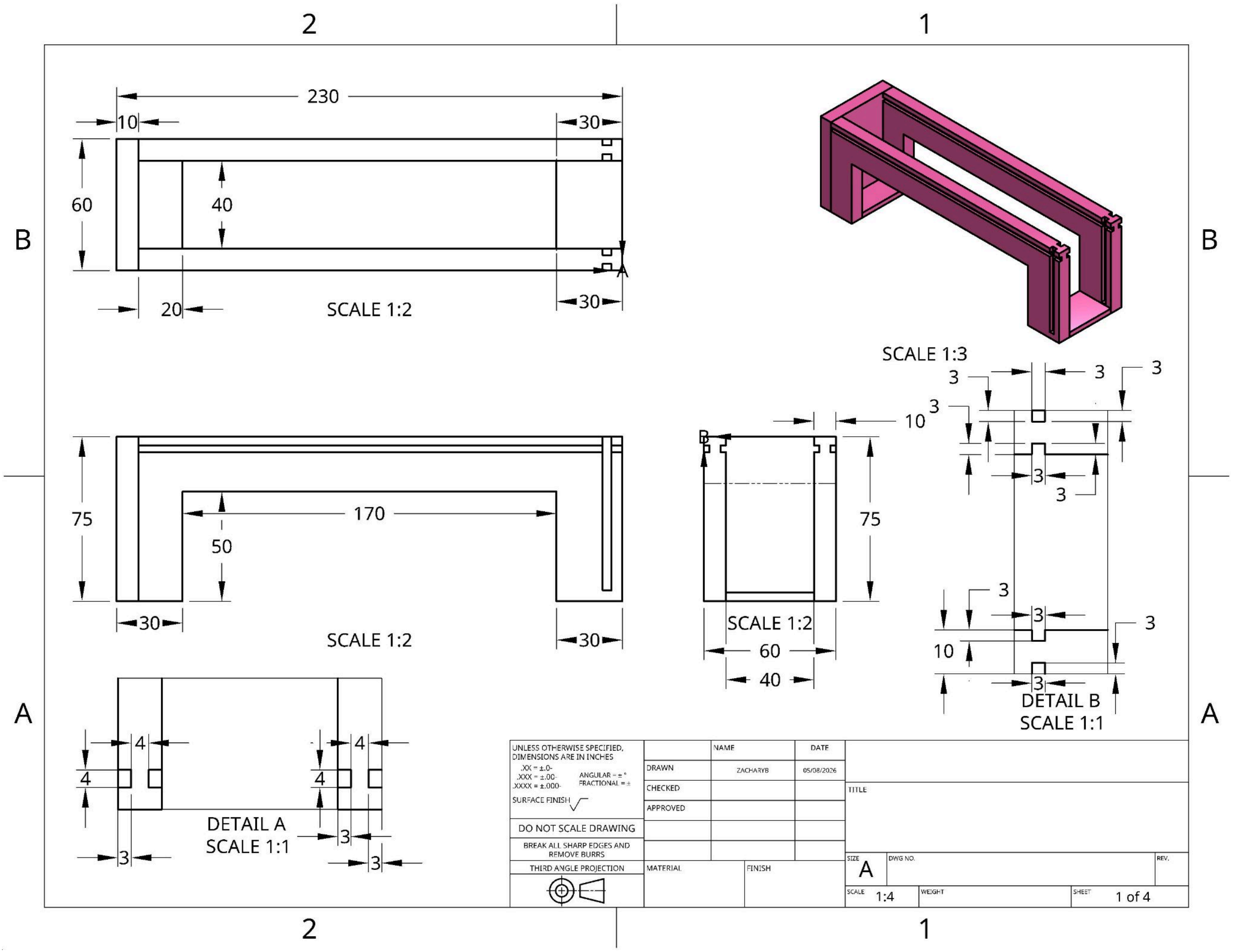
(Front wall slider)



(top slider)



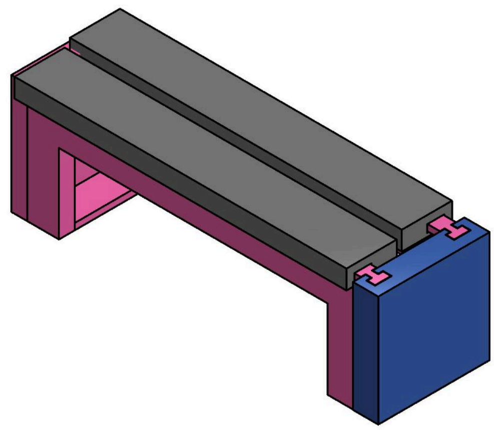
(Main body)



(Full assembly)

2 1

B B



SCALE 1:2

A A

DWG NO.	SIZE	SCALE	SHEET	REV.
	A	1:4	4 of 4	

2 1

	Proposed Solutions					
Design Requirements	**Current Solution: Crawling under the table to plug in the cable**	**Idea #1: 3D printed power strip cover**	**Idea #2: Wall plug clip**	**Idea #3: Velcro**	**Idea #4: Magnets**	**Idea #5: Spiral Cable**
Safe to use-High 2 points	0	+2	+2	+2	-1	+1
Doesn't permanently damage outlets-High 1 points	0	+1	-1	+1	-1	+1
Has to be less than $0.50 per outlet or cable secured-High 2 points	0	-1	-1	0	-1	-2
Durable enough for at least 5 installments-High 1 points	0	0	+1	-1	+1	+1
Can be used on a large variety of outlets and cables-high 2 point	0	0	-2	+2	+2	+2
Has to be easily implemented, can be installed in under 60 seconds-medium 1 point	0	+1	+1	+1	-1	+1
Has to be easy to use for a child over the age of 10-medium 1 point	0	+1	0	+1	+1	+1
Total Score	**0**	**4**	**0**	**7**	**0**	**5**